RISHIK THAMMISETTY

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SUMMARY

Third-year Electrical and Electronics Engineering undergraduate at NIT Andhra Pradesh with interests in Machine Learning, VLSI, and Electronic Design Automation. Currently working as a Research Intern at IIT Bhilai on deep learning for image inpainting while developing physics-informed AI solutions for semiconductor applications.

EDUCATION

National Institute of Technology Andhra Pradesh — **Aug 2024 – Present**
B.Tech. in Electrical and Electronics Engineering | CGPA: 8.35/10 (Current) — Tadepalligudem, India

National Institute of Technology Andhra Pradesh — **Aug 2025 – Present**
Minor Degree in AI and Robotics

PROFESSIONAL EXPERIENCE

Indian Institute of Technology (IIT) Bhilai — **May 2026 – Present**
Research Intern

- Working as a research intern on image inpainting, focusing on developing improved deep learning-based models for realistic image reconstruction.
- Exploring convolutional and attention-based architectures to enhance inpainting quality on corrupted or masked image regions.

National Institute of Technology Andhra Pradesh — **Apr 2025 – Present**
Volunteer – E-Yantra Club

- Coordinated technical events and supported hands-on training sessions.
- Developed leadership, teamwork, and problem-solving skills through collaborative club activities.

TECHNICAL SKILLS

Languages : C, Python, Matlab, Verilog

ML Frameworks & Tools : PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, SHAP

Core Domains : Machine Learning, Deep Learning, Image Processing, VLSI Design, EDA

Other Tools : Power BI, Web Development, ESP32, Embedded Systems

ACADEMIC PROJECTS

Image Inpainting – IIT Bhilai Research Project — **May 2026 – Present**

- Developing a deep learning-based image inpainting system in association with IIT Bhilai to reconstruct missing or corrupted image regions with high visual fidelity & Conducting performance evaluation using standard image quality metrics and optimizing model architectures for improved reconstruction accuracy and efficiency.

Physics-Informed ML for CMOS Gate Delay Prediction — **May 2026 – Present**

- Developed a physics-informed deep learning framework for **45nm CMOS gate delay prediction** by integrating semiconductor physics with PyTorch-based neural networks.
- Applied SHAP-based explainability techniques to validate that the model prioritizes physically meaningful semiconductor parameters for delay prediction.

EV Battery Health Prediction — **Jan 2026 – April 2026**

- Developed an integrated ML-based system to predict State of Health (SoH) using SoC, RUL, and anomaly detection as supporting features. & Applied data preprocessing, feature engineering, and regression techniques to improve battery degradation prediction accuracy under varying operating conditions.

Smart Solar Street Light System — **Feb 2026 – April 2026**

- Designed an IoT-based adaptive lighting system using ESP32 with traffic-based brightness control and energy monitoring and optimization.